SafeLens Montgomery

Community Safety Intelligence Platform

World Wide Vibes Hackathon | Public Safety, Emergency Response & City Analytics
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Bright Data	GenAI Works	City of Montgomery

TEAM

Name	GitHub	Email
Angela Kwok (Team Lead)	github.com/angelatyk	angelatyk@gmail.com
Andres Linero	github.com/andres-linero	andres.nyc.col@gmail.com
Varoun Bajaj	github.com/varounbajaj-proj	varounbajaj@gmail.com
Robert K. Cummings Jr	github.com/Dadillacrob1001	RobKCJr010210@gmail.com

Live Site	Demo (With Demo Accounts)	Repository
safelens-montgomery.vercel.app	safelens-montgomery.vercel.app/demo	github.com/angelatyk/safelens-montgomery

1. The Problem

Montgomery, Alabama carries an extraordinary history — birthplace of the Civil Rights Movement, home of Dexter Avenue Baptist Church, the Bus Boycott, and the Selma March. It is a city that has always understood the relationship between community trust and public institutions. Today, that trust is under pressure from a public safety system stretched beyond its capacity.

The police department is significantly understaffed — officers cannot be everywhere, dispatchers manage emergencies alongside a flood of non-emergency calls on the same 911 queue, and residents are left with fragmented, decontextualized information. City leadership hears about public safety constantly in city hall, yet the tools to act on it in real time remain limited.

Pain Point	Impact
Understaffed police department	Incidents go undetected or under-prioritized without AI assistance
911 queue overload	Non-emergency calls lengthen hold times for genuine crises
No AI force multiplication	Officers cannot anticipate threats or intelligently prioritize responses
Residents lack context	Raw data creates fear without understanding; community disengaged
No two-way communication	No channel for proactive alerts or for residents to report to officials
Data without responsible framing	Crime maps create alarm, not clarity — no narrative context provided

2. The Solution

SafeLens Montgomery is a bidirectional community safety intelligence platform purpose-built for this city — not a generic dashboard dropped on top of a GIS feed. It ingests data from six real-time sources, runs them through an AI pipeline that clusters and validates events across sources, and surfaces human-readable safety narratives that help officials and officers prioritize their resources where they are needed most. Residents, in turn, can report and confirm incidents directly — staying more aware of what is happening around their city while actively contributing to the picture officials rely on.

MISSION	*Replace raw incident numbers with responsible, context-rich safety narratives — empowering residents to stay aware, enabling officials to prioritize resources effectively, and amplifying an understaffed department through AI-driven intelligence.*

The core thesis: AI bridges the staffing gap. When AL.com, WSFA, and a community report all reference the same incident, the pipeline detects that convergence, scores it for credibility, and surfaces a validated event — making a lean team operate with the awareness of a much larger one.

AI Capability	How It Multiplies Force
Cross-Source Validation	NLP pipeline clusters articles from 6 sources by text similarity and time window — only events corroborated by multiple sources become incidents
Credibility Scoring	Each cluster scored 0–100: source count (up to 60 pts), source quality (up to 25 pts), severity (up to 15 pts). Only clusters scoring 25+ enter the database
AI Narrative Generation	Claude Sonnet synthesizes validated incidents into plain-language safety updates with estimated resolution times — pulled from incidents, news articles, and resident reports
Narrative-Driven Dispatch	Officials review AI narratives alongside source evidence and live polling to prioritize where to send officers — replacing manual data-sifting with structured decision support

3. Data Sources & Intelligence Pipeline

SafeLens aggregates from six real-time data sources, processes them through a multi-stage NLP pipeline, and stores validated incidents in Supabase. All data is cross-referenced — no single source creates an incident unilaterally.

Source	Contents	Volume / Cadence
RSS Feeds (AL.com, WSFA)	Local news articles from Montgomery's primary outlets via Bright Data SERP	Continuous
Google News	Broader Montgomery safety coverage — scored before insert; below 0.15 relevance discarded	400+ articles collected
Twitter / X	Real-time community signals, weighted lower than institutional sources	Real-time
311 Service Requests	Montgomery's official non-emergency service request database	279,000 reports
Official Alerts	Breaking alerts from Montgomery PD, NWS, and local TV stations	Real-time
Neighborhood Boundaries	Census GeoJSON polygons used to geo-tag all incidents by neighborhood	Static

NLP Pipeline — Collect → Validate → Narrate

Stage	What Happens
1 — Classify	12 incident types × 4 severity levels (critical / high / medium / low)
2 — Cluster	TF-IDF similarity + time window groups articles about the same event
3 — Score	0–100 credibility: source count (60 pts) + quality (25 pts) + severity (15 pts). Min 25 to enter DB.
4 — Incidents	High-scoring clusters become validated incidents, cross-linked via junction tables
5 — Narratives	Claude Sonnet synthesizes incidents + news + resident reports into plain-language updates
6 — Feedback	15-min community polling auto-resolves stale incidents; source trust adjusts dynamically

4. Features Built

Seven core features were fully implemented and are live in the deployed application.

Feature	What Was Built
Data Ingestion Pipeline	Python workers collect from all 6 sources. Google News scored before insert — below 0.15 relevance discarded. NLP pipeline auto-triggers narrative generation after clustering for an end-to-end automatic flow.
Incident Card Feed	Cards display the AI narrative, generation timestamp, and model used. Official status visible (acknowledged / verified / dispatched / resolved) with any public note. Community polling every 15 min — enough "no longer active" votes auto-resolves the incident.
Resident Report Submission	Floating "Report Something" button always visible. Frictionless form — category, description, optional location, no login required. Reports feed into the NLP pipeline as a narrative source.
Verified by City Badge	Live status badge (Submitted → Verified → Resolved) updated in real time via Supabase Realtime. Officials attach a public response note that renders on the card.
Official Narrative Queue	Auth-gated dashboard. Each narrative shown with its source breakdown (e.g. "1 Incident, 2 News Articles"), live polling results, and resident comments (UI built). Officials use this to prioritize dispatch — structured decision support, not raw data.
AI Safety Narratives (Claude Sonnet 4)	Claude synthesizes validated incidents into warm, community-oriented plain-language updates from 3 source types. Estimates resolution times. SHA-256 hashing prevents duplicates. Cross-linked to sources via junction tables.
Public Safety News Feed	Bright Data SERP queries local sources (AL.com, WSFA). Relevance threshold 0.3; social media and noise keywords excluded. Headlines shown with AI summaries. Sources API available for drill-down.

5. Architecture & Technology

LAYER 1 — DATA INGESTION (Python Workers)
RSS Feeds | Google News | Twitter/X | 311 Reports (279K) | Official Alerts | Resident Submissions

↓ **NLP Pipeline: classify → cluster (TF-IDF) → score → incidents → narratives → feedback** ↓

LAYER 2 — AI ANALYSIS
Gemini 2.0 Flash: classification, clustering, hotspot detection | Claude Sonnet 4: narrative generation, responsible framing

↓ **Validated incidents + narratives stored in Supabase PostgreSQL** ↓

LAYER 3 — API (Next.js API Routes)
/incidents /narratives /neighborhoods /alerts /news /submit /acknowledge /resolve /broadcast

↓ **Served to frontend via Supabase Realtime for live updates** ↓

LAYER 4 — FRONTEND (Next.js 14 + Tailwind + shadcn/ui)
Resident View: Incident Feed + Report Submission + News Feed | Official View: Narrative Queue + Source Evidence + Dispatch Prioritization

Technology	Purpose
Next.js 14 (App Router) + TypeScript	Full-stack framework — frontend and API routes
Tailwind CSS + shadcn/ui	UI component system
Supabase (PostgreSQL + Auth + Realtime)	Database, authentication, live push updates
Bright Data SERP API	Local news scraping (AL.com, WSFA) — all 4 Bright Data tools integrated
Python (NLP pipeline + workers)	Data ingestion, TF-IDF clustering, credibility scoring, auto-trigger narrative flow
Gemini 2.0 Flash	Fast classification of incident types and severity levels
Claude Sonnet 4 (Anthropic)	On-demand responsible safety narrative generation from validated incidents
Vercel + Railway	Frontend deployment + Python worker hosting

6. AI Strategy & Responsible Design

SafeLens uses a dual-model AI architecture. Gemini 2.0 Flash handles fast, high-volume classification on every ingestion cycle. Claude Sonnet 4 handles quality-sensitive, responsible narrative generation on demand — and is explicitly prompted to avoid stigmatizing language and to ground every statement in verified data.

PROMPT PATTERN	*"You are a responsible public safety communicator for the City of Montgomery. Given: [validated incidents, news articles, resident reports], generate a community-oriented safety update that is factual, context-aware, estimates resolution time, and avoids language that could stigmatize residents. Data: {context}"*

Responsible Design Decision	Implementation
Multi-source validation	No single source creates an incident — corroboration required. Credibility threshold enforced at 25/100.
Noise filtering	Google News scored at 0.15 threshold; social media excluded from narrative sources; noise keywords purged
No stigmatization language	Claude explicitly prompted against stigmatizing language; warm, community-oriented tone required
Source transparency	Every narrative cross-linked to source articles and incidents. Officers see exact source counts in the detail panel.
Community feedback loop	15-min polling lets residents validate or invalidate active incidents — platform adjusts source trust over time
Context over raw counts	Incidents shown with trend context and resolution time estimates — never as standalone alarming statistics

7. Database Schema

SafeLens uses a 12-table PostgreSQL schema in Supabase. The schema reflects the platform's two core flows: the AI pipeline (ingestion → incidents → narratives → feedback) and the community loop (resident reports → official review → public resolution).

AI PIPELINE	COMMUNITY LOOP	SUPPORTING TABLES
`incidents` Source-tagged events from pipeline `news_articles` Scored articles (threshold 0.15+) `narratives` AI-generated safety updates `narrative_incidents` Narrative ↔ incident links `narrative_news_articles` Narrative ↔ article links `source_reputation` Dynamic source trust scores	`resident_reports` Geolocated community reports `narrative_feedback` Accurate / not-relevant votes `narrative_comments` Resident comments on narratives `narrative_public_updates` Official public response notes	`neighborhoods` GeoJSON polygons + safety scores `users` Auth + roles (resident / official) `alerts` Official broadcast alerts `weather_context` Daily weather for spike correlation

Key relationship: `narratives` is the central table — each narrative links back to its source `incidents` and `news_articles` via junction tables, carries an `official_status` lifecycle (unreviewed → acknowledged → verified → dispatched → resolved), and accumulates `narrative_feedback` from community voting and `narrative_public_updates` from officials.

8. Impact & Future

SafeLens was built specifically for Montgomery — its history, its demographics, its understaffed department, its open datasets. The Civil Rights Movement was powered by community information networks; SafeLens is a modern expression of that same principle: when residents and officials share a common, accurate picture of what is happening, trust is built and action becomes possible.

Operationally, the platform multiplies a lean department's capacity through AI-driven triage and narrative decision support. The 279,000 Montgomery 311 records, real-time multi-source pipeline, and community polling feedback loop produce the kind of real-time intelligence that can be taken directly to the Mayor for decision-making. Commercially, Montgomery is the pilot for a SaaS platform deployable to any mid-sized U.S. city — long-term, SafeLens expands into a full municipal communication platform across city services, infrastructure, workforce development, and civic engagement.

Future Improvements

Feature	Description
Official Alert Broadcast	City-wide alerts (Info / Warning / Emergency) pushed live to all resident sessions via Supabase Realtime
Interactive Map View	Mapbox heatmap for incident density, resident pins, neighborhood polygons with click-to-narrative
Weather & Events Integration	Already in schema (weather_context table) — wire NWS data + event calendar to auto-tag incident spikes
Resident Comments	Schema built (narrative_comments) — wire frontend comment threads to narrative cards
Trend Analytics	7/30/90-day incident trend charts; news volume vs. 911 volume correlation view
Map Directory	Searchable neighborhood resources, city contacts, and services layered on the safety map
Full Municipal Platform	Expand beyond public safety: city services, infrastructure reporting, civic engagement, community programs

VISION	*SafeLens Montgomery is proof that a small team, real data, and responsible AI can equip a city with powerful, advanced tools. The goal beyond this hackathon is to make Montgomery the model for how mid-sized American cities use AI to close the gap between the people they serve and the officials who serve them.*

SafeLens Montgomery

Making Montgomery's communities safer through data, context, and connection.